

Chapter 1

Poker Without Revealing the Cards

1.1 Objective

This document accompanies the executable poker demonstrator and presents the same construction in a more formal order. The implementation does not yet attempt to solve the full game of Texas Hold'em; it isolates the dealing problem into four stages that can be inspected in code and in runtime logs.

1.2 Scientific Scope

The present demonstrator addresses the following narrow question:

Can mutually distrustful parties instantiate a poker hand without a trusted dealer while preserving privacy of hole cards until showdown?

The current answer is partial rather than complete. The demonstrator models:

1. randomization of a canonical deck,
2. secure multi-party shuffling,
3. secure dealing of private and public cards,
4. public commitment to private hands.

It does not yet model betting, hand-ranking evaluation, or adversarial network execution.

1.3 Source Mapping

- C source directory: `/src/src/poker_without_revealing_the_cards/`
- Primary implementation file: `poker_without_revealing_the_cards.c`
- Build entrypoint: `demostrations/Makefile`
- PDF helper: `./book -pdf section_poker`

1.4 Protocol Decomposition

Stage 1: Deck Randomization

A standard 52-card deck is permuted and a public digest of the resulting state is published. The digest fixes the transcript without revealing the deck order.

Stage 2: Secure Multi-Party Shuffling

Each player privately shuffles the deck and applies a commutative-encryption layer. After each transformation, the encrypted deck is committed to publicly. Once the final encrypted deck is fixed, the demonstrator also publishes a hash for each encrypted card that will later be dealt. These public hashes are the visible form of the sampled cards during this stage.

Stage 3: Secure Dealing

Hole cards are privately recovered by removing encryption layers in a player-specific order. Community cards are jointly decrypted and become public. Before each decryption, the demonstrator verifies that the drawn encrypted card matches the Stage 2 hash commitment for that deal slot. It also validates that no plaintext card appears twice.

Stage 4: Hand Commitments

If player P_i holds hole cards $h_{i,1}, h_{i,2}$, the player samples a nonce N_i and publishes

$$C_i = H(N_i \parallel 2 \parallel h_{i,1} \parallel h_{i,2}).$$

Later, N_i and the private cards are revealed so that every participant can recompute C_i .

1.5 Build and Run

```
cd /src
make -C demonstrations demo_poker
```

Alternative suit rendering modes:

```
make -C demonstrations demo_poker POKER_ARGS="--suits ascii"
make -C demonstrations demo_poker POKER_ARGS="--suits unicode"
POKER_SUIT_MODE=unicode make -C demonstrations demo_poker
```

1.6 Observed Runtime Structure

The program emits a staged trace of the form:

```
Stage 1  Randomization of the deck
Stage 2  Secure multi-party shuffling
Stage 3  Secure multi-party dealing
Stage 4  Hand commitments and validation
```

The log also records public commitments, encrypted deck previews, dealt cards, sampled-card hash commitments, sampled-card verification, and commitment verification at showdown.

1.7 Limitations

The current demonstrator should be interpreted as a protocol sketch with an executable transcript, not as a final cryptographic poker engine. Open work includes:

1. betting rounds and action synchronization,
2. formal hand evaluation,
3. authenticated transport,
4. malicious-security guarantees,
5. stronger proof obligations for correct shuffle and deal operations.